

LensWalk: Agentic Video Understanding by Planning How You See in Videos

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Abstract

The dense, temporal nature of video presents a profound challenge for automated analysis. Despite the use of powerful Vision-Language Models, prevailing methods for video understanding are limited by the inherent disconnect between reasoning and perception: they rely on static, pre-processed information and cannot actively seek raw evidence from video as their understanding evolves. To address this, we introduce LensWalk, a flexible agentic framework that empowers a Large Language Model reasoner to control its own visual observation actively. LensWalk establishes a tight reason-plan-observe loop where the agent dynamically specifies, at each step, the temporal scope and sampling density of the video it observes. Using a suite of versatile, Vision-Language Model based tools parameterized by these specifications, the agent can perform broad scans for cues, focus on specific segments for fact extraction, and stitch evidence from multiple moments for holistic verification. This design allows for progressive, on-demand evidence gathering that directly serves the agent’s evolving chain of thought. Without requiring any model fine-tuning, LensWalk delivers substantial, plug-and-play performance gains on multiple model recipes, boosting their accuracy by over 5% on challenging long-video benchmarks like LVBench and Video-MME. Our analysis reveals that enabling an agent to control how it sees is key to unlocking more accurate, robust, and interpretable video reasoning.

1. Introduction

Video understanding is a core task in computer vision with profound challenges, given the multi-layered, high-density, and temporally unfolding nature of its information [13, 45, 49]. This volume overwhelms the finite cogni-

tive resources of any analytical agent, be it biological or artificial, making uniform, exhaustive analysis intractable and inefficient [1, 5]. Powerful intelligence, like humans, cope through purposeful information seeking [1, 5, 34]: we continually adapt—shifting from broad peripheral scans to sharp foveal focus—and intermittently reflect and verify, guided by continual reasoning that evolves both intent and observational priorities. We contend that this principle is key to reliable, accurate video analysis: observation should be actively scheduled under limited capacity. However, realizing such purpose-oriented video observation planning in practice remains an open challenge.

While the progress of Vision-Language Models (VLMs) [2, 18, 23] and test-time thinking [15, 29] has significantly advanced the field, this principle is far from realized in current video understanding pipelines. In typical setups, long videos are sampled within context windows and converted once into a single context; this uniform strategy often drops sparse but decisive events or drowns them in redundancy [24, 41, 49]. More recent efforts instead try heuristic frame or token selection [24, 36, 47, 58] to preserve “key” information under a constrained context length, but these heuristics still rely on fixed, once sampling, cannot shift focus or reallocate budget as intermediate hypotheses change, and thus struggle to support the stepwise resolution of complex questions.

Beyond such single-model setups, many recent retrieval-based and agentic frameworks [26, 32, 43] explicitly try to expand information as reasoning unfolds. Reactive retrieval pipelines [28, 35, 69] let the model trigger tools to fetch pre-processed artifacts such as ASR transcripts, OCR, or clip-level captions when new questions or intermediate hypotheses arise; however, perception is still bound to a largely fixed, hand-crafted workflow and a predetermined menu of views on the video, with limited ability to decide granularity or revisit the raw video when ambiguity appears.

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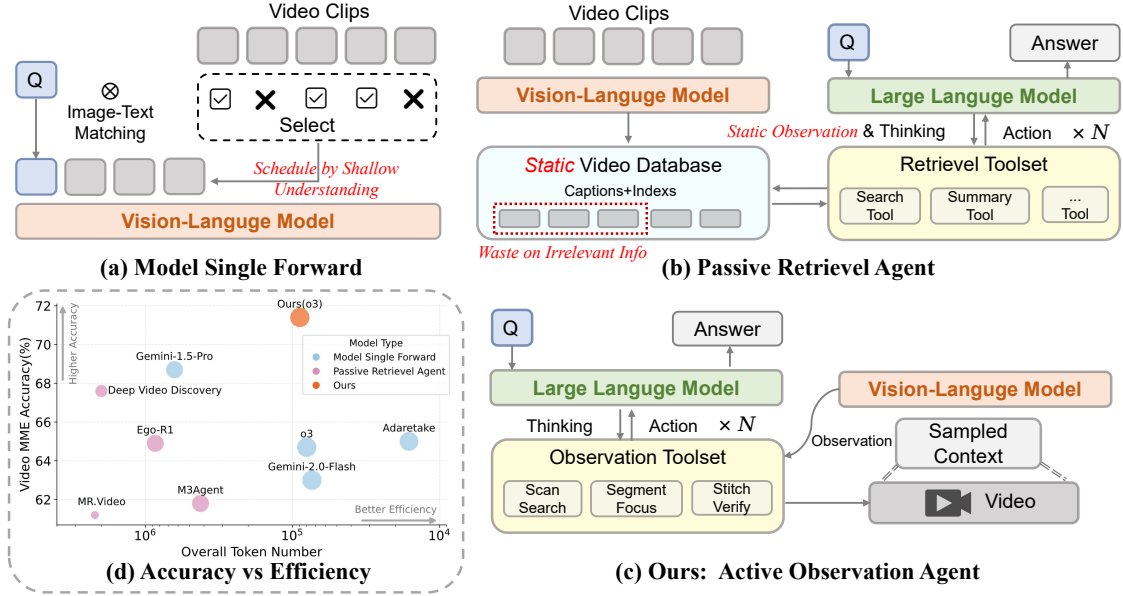


Figure 1. Comparison of video understanding paradigms: (a) Single Forward of model with one-off context selection. (b) Retrieval-Based Video Agent. (c) **LensWalk**, an agent framework that actively plans observations to fulfill subsequent reasoning, explores iteratively, and self-regulates its observation budget. (d) By actively planning observations only on video segments essential to its reasoning process, LensWalk simultaneously achieves high accuracy and exceptional efficiency.

Video agents with tool flows [26, 43, 63], as well as those that let an LLM controller query over pre-computed video indexes, offer more flexible querying but still operate on static representations rather than scheduling fresh observations from the source video, so observation and reasoning remain only loosely coupled and the overall budget remains underplanned.

To this end, we present **LensWalk**, an agentic framework that gives the LLM reasoner on-hand control over where, and how dense to look at the video for active observation. LensWalk organizes inference as a multi-round *reason-plan-observe* loop: at each round, the reasoner reflects on the current question and accumulated evidence, then plans an observation action by selecting an observation tool and its parameters, which determine the visible context—which temporal regions of the source video to inspect and at what sampling level. This loop is instantiated by a suite of complementary observation tools powered by a vision-language model that converts the prepared context into visual evidence: *Scan Search* supports parallel sweeps over broad, disjoint candidate ranges for cue discovery, *Segment Focus* constructs compact, detail-oriented views for a targeted interval, and *Stitched Verify* assembles custom samples from multiple key segments into a single context for global checking. Each tool is encouraged to return frame-interleaved timestamps aligned with discovered key information as anchors for subsequent rounds. As observations are acquired progressively from raw video, LensWalk main-

tains a round-updated global cast table as lightweight hot memory, tracking entities and their relations across rounds without repeatedly re-reading long histories. By tying each observation to an explicit, parameterized plan, LensWalk lets the agent schedule where to look and how densely to sample the video directly from its evolving reasoning state, reallocating its frame budget across regions and time scales as hypotheses change.

Our empirical results on long video understanding and video reasoning benchmarks reveal that this framework is not only conceptually appealing but also practically effective. As a plug-and-play agentic framework, LensWalk consistently improves even strong Reasoner–Observer combinations; when a single model serves as both to make self-scheduled observation, it effectively offers a “free lunch”, yielding substantial gains—for example, the powerful o3 model improves by over 11.5% and 6.7% accuracy on challenging long-video benchmarks such as LVBench and VideoMME (long split). At the same time, LensWalk exhibits an emergent ability to manage observational cost: it adaptively resolves easy queries with minimal turns, while allocating more budget only to genuinely uncertain cases, as in Figure 1(d), leading to total token usage far below that of retrieval agents that consume millions of tokens and only slightly above a single dense pass over long frame sequences. Analysis of its reasoning traces further reveals diverse, human-like strategies—strategic reflection, progressive zoom-in, and integrative verification—replacing the

fixed workflows or degenerative action repetition common in prior agents, and confirming that the performance and efficiency gains stem from its core principle of actively planning observation guided by reasoning.

2. Related Work

Context Selection on Video Understanding The constrained context window of Multimodal Large Language Models (MLLMs) has motivated a line of work on selecting a compact yet informative video context. Existing approaches typically either perform query-aware keyframe or clip selection with temporal and semantic diversity, or learn task-specific alignment modules to pick task-relevant snippets [3, 16, 24, 33, 41, 58, 60]. Another line focuses on information condensation by pruning redundancy at frame, clip, or model-layer granularity [36, 37, 47, 56, 65]. While these strategies improve the utilization of the limited token budget, they still choose the visible context in a single shot before reasoning, whereas our agent treats context selection itself as a step-wise, tool-mediated process driven by its evolving hypotheses.

Test-Time Scaling on Video Understanding Recent advances in Test-Time Scaling (TTS) for video understanding allocate additional computation at inference time to improve performance. One line of work strengthens internal reasoning by training models to support long, self-corrective chains of thought with verifiable rewards or preference-based objectives, as in Video-R1 and VideoRFT [12, 15, 48], but still operates on fixed visual inputs. A complementary line wraps video models in agentic workflows, where LLMs control how to retrieve or aggregate preprocessed clips and captions: building on early pipeline-based systems that query clips, frames, and metadata [35, 40, 46, 69], recent works introduce reasoning-based agents that schedule clip retrieval and caption aggregation over long videos [26, 32, 43, 62]. In contrast to both paradigms, our framework scales test-time inference by letting the agent explicitly decide *how* to observe the video—choosing temporal ranges, sampling densities, and segment compositions—rather than only allocating more computation over a fixed observation stage.

Multimodal Tool-Use Agents Tool use is a core capability of agents, enabling models to proactively invoke external tools and thereby extend their effective context beyond intrinsic parametric knowledge [11, 52, 62]. Recent work on multimodal tool-use agents largely follows this perspective, but instantiates it along several directions. One prominent line is agentic search [6, 20, 21, 25, 44, 50, 53], where agents actively query textual and visual sources to retrieve evidence that augments their internal reasoning

context. A complementary line is agentic visual processing [4, 19, 38, 39, 51, 54, 61, 66, 68, 70], which equips agents with tools that transform, compose, or generate images during problem solving. For video understanding, recent tool-based methods usually interface with video through preprocessed surrogates: they convert videos into databases of dense captions or clip embeddings, design indexing schemes, and then let the agent query these static repositories with text-based tools [26, 43, 62]. In such systems, the agent can decide *what* to ask but has little control over *how* the video is observed; the temporal coverage, sampling density, and composition of segments are largely fixed by the preprocessing pipeline or a small set of coarse tool options.

3. Method

Despite recent great progress, most video understanding pipelines still pair strong reasoners with a fixed observation stage, so the scope and granularity of visual evidence do not adapt to the agent’s evolving thoughts. We instead view video understanding as also a problem of deciding how to observe, and LensWalk is designed precisely to let an agent dynamically choose where to look and at what granularity. see Appendix A for pseudocode and concrete prompts.

3.1. Overview of the LensWalk Architecture

LensWalk models video understanding as an active observation process, driven by an agent that intelligently decides **where to look, what to observe, and how densely to sample** the video timeline per reasoning step. At its core, an LLM-based **Reasoner** (M_r) analyzes the user’s query and accumulated evidence to formulate the next plan by calling observe tools with controllable video scopes. These tools belong to a versatile **Observation Toolkit** ($\mathcal{O}$), which leverages a core VLM as its **Observer** (M_o) to extract information directly from a specific visible video context according to the Reasoner’s arguments. To ensure coherence across multiple turns, *Timestamp Anchors* and a global *Subject Memory Table* are designed as lightweight memory mechanisms that provide evidence grounding. This establishes an iterative reason-plan-observe cycle. In the following subsections, we sequentially introduce how the agent schedules visual observations during its multi-turn tool use, the design of the observation toolkit, and the memory grounding mechanisms that ensure multi-turn coherence. Figure 2 illustrates the LensWalk agent pipeline on a real video understanding instance. In the following subsections, we sequentially introduce how the agent schedules visual observations during its multi-turn tool use, the design of the observation toolkit, and the evidence grounding mechanisms that ensure multi-turn coherence.

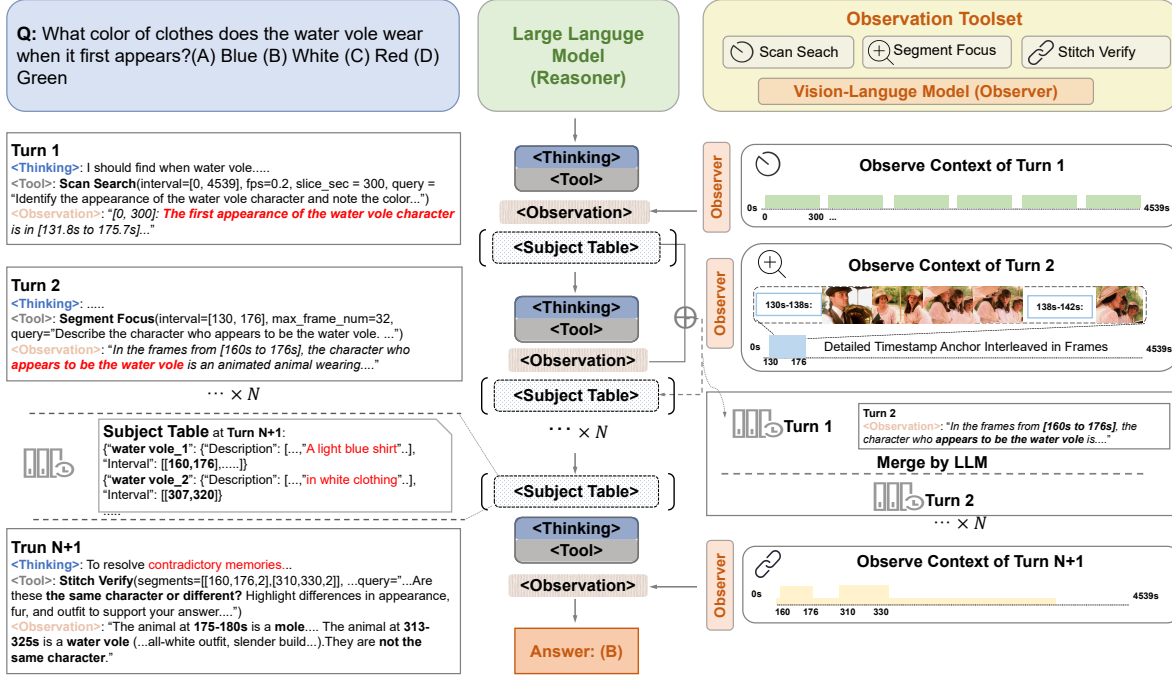


Figure 2. Illustration of LensWalk’s reasoning-scheduled active observation on a real trace. The LLM reasoner alternates Scan Search and Segment Focus to seek question-relevant evidence, initially misclassifies a mole as the queried water vole and records conflicting memories in the subject table (red text), then invokes Stitch Verify to inspect key segments, resolve the contradiction, and arrive at the correct answer.

3.2. Reason-Scheduled Video Observation

To flexibly acquire evidence, the LensWalk agent schedules its visual observation by parameterizing each tool call based on its ongoing reasoning. At each step t , the **Reasoner** (M_r) integrates the user query Q , video metadata V_{info} (e.g., duration, camera perspective), and the accumulated history of prior actions and observations L_{t-1} to produce a fully specified plan, a_t .

$$a_t = M_r(Q, V_{\text{info}}, L_{t-1}).$$

This plan $a_t := (o_t, q_t, \mathcal{I}_t, \rho_{o_t})$ is a tool call that precisely decides the next observation’s scope and budget, as detailed below:

$$\begin{cases} o_t \in \mathcal{O} & \text{The observation tool to invoke} \\ q_t & \text{The guiding sub-question} \\ \mathcal{I}_t & \text{Scope and Budget controls (e.g., temporal interval(s), frame count/FPS)} \\ \rho_{o_t} & \text{Tool-specific arguments} \end{cases}$$

Executing this plan samples the specified frames (according to $\mathcal{I}_t$ and ρ_{o_t}) and, together with q_t , feeds them to the **Observer** (M_o), which returns evidence e_t as a new observation. The plan-observation pair is then appended to the history as a new conversation turn: $L_t = [L_{t-1}, \{a_t, e_t\}]$. The explicit inclusion of controllable scope parameters $\mathcal{I}_t$ in each action is critical. It transforms the history L_t into a structured document with precise temporal tags. Furthermore, this history of parameterized calls allows the agent

to track its own exploration progress, enabling it to target under-explored video sections in subsequent steps.

3.3. Multi-Granularity Observation Tool Suite

The Reasoner’s plan is executed by a compact observation toolkit. Designed to enable diverse compositions of observation contexts, it offers a minimal yet expressive toolset: *Scan Search* (broad localization), *Segment Focus* (fine-grained inspection), and *Stitched Verify* (cross-segment integration), plus a *Finish* action. This effectively supports the reasoning-driven behaviors analyzed in Section 4.4. Figure 2 illustrates their arguments and examples.

Scan Search. This tool is designed for efficient, broad discovery across a specified temporal interval. In addition to a guiding sub-question (q_t) and scope controls ($\mathcal{I}_t$), the tool-specific settings (ρ_{Scan}) require a partitioning strategy, defined by either the number of slices (n_s) or the duration of each slice (d_s). The tool partitions the given interval, sparsely samples frames from each slice, and queries the VLM on a slice-by-slice basis. It returns per-slice summaries or relevance judgments that indicate the likely locations of target events. As each slice is processed independently, this operation is inherently parallelizable. This makes it well-suited for an initial exploration of still unfamiliar intervals or for detecting the presence of object or event cues, as its parallel nature allows for efficient localization of potentially relevant sections.

Segment Focus. This tool executes a focused probe over a single, continuous time interval, applying a dense sampling strategy as defined by the scope controls ($\mathcal{I}_t$) to answer a precise sub-question (q_t). It is designed to extract fine-grained details, local dynamics, or small visual attributes that a coarser observation might miss. This operator is therefore ideal for hypothesis verification, attribute reading, or disambiguating similar actors and events. It also allows for re-examining the same interval with an even denser sampling configuration to clarify previously indistinct observations, a flexibility not easily available in methods reliant on pre-embedded video clips.

Stitched Verify. This tool is designed to observe facts that require integrating evidence from multiple, non-contiguous video segments, such as for before-and-after comparisons, tracking object transfers, or verifying causal narratives. Its tool-specific settings, ρ_{Stitch} , take a list of segments $S = [(s_i, e_i, r_i), \dots]$, where each entry defines a segment by its start time (s_i), end time (e_i), and a specific sampling rate for the segment (r_i). The operator consolidates frames from these distinct intervals into a single, coherent batch for the VLM. This uniquely supports asymmetric sampling—allocating a higher frame rate to action-heavy segments while using a lower rate for transitional ones. This enables the agent to verify a complex chain of events in a single step, reasoning over a unified visual context rather than reconciling isolated observations solely in the language domain.

3.4. Evidence Grounding for Long-Horizon Coherence

Because LensWalk runs multiple observation turns at different scopes, it must reconcile potentially ambiguous evidence across them. We therefore introduce two lightweight grounding components—*Timestamp Anchors* and a *Subject Memory Table*—that tie observations to precise temporal ranges and globally consistent entity references.

Timestamp Anchors within Tool Observation. During each observation, as in Figure 2, detailed timestamp anchors at a specified granularity would be inserted to *Observer* its visual context (frame sequences). These anchors are inserted directly within the tool’s visible frames as text, prompting the *Observer* to ground its answers with explicit temporal references (e.g., “at 01:15-01:40”). This provides the *Reasoner* with fine-grained temporal cues, enabling it to efficiently pinpoint evidence sources in subsequent planning cycles, schedule re-observations over ambiguous ranges, and prune irrelevant sections for more precise analysis.

Subject Memory Table for Entity Consistency. After each observation, as in Figure 2, a global *Subject Memory Table* Sub_t is updated by an LLM with entity information

from the new tool observation. It maintains outside the main reasoning history L_t , records persistent entities—such as people or objects—along with summarized attributes and seen timestamps. Before each reasoning turn, it would be inserted into the context and overwrite the previous version. This evolving memory serves a dual purpose: it allows the Reasoner to use canonical identifiers for efficient, consistent entity reference, eliminating the need to repeatedly re-identify entities from the tool history in each reasoning cycle. It also provides a clean, structured knowledge base that the Reasoner consults to strategically plan its subsequent observations. This mechanism manages accumulating entity information to maintain cross-round reference stability, and avoids redundant context tokens from lengthy descriptions, ensuring the understanding of subjects remains consistent as it progressively explores the video.

Maintaining consistent, reliable memory across multiple reasoning turns is challenging in agentic video understanding [26, 32, 62], and prior systems often resort to heavy pre-scanning, summarization, or intricate bookkeeping schemes [26, 43, 46, 62], which incur considerable computational and engineering overhead. Our evidence grounding instead maintains a compact, structured state and, by tying evidence to temporal scopes and entities in the history, allows the agent to revisit ambiguous regions via re-observation, as illustrated in Figure 2. Although Observer or memory-updating errors can still introduce inconsistencies, this lightweight combination of grounding and re-observation yields a scalable, self-corrective pipeline that fits into the reason–plan–observe loop and is supported by the ablations in Section 4.3.

4. Experiments

4.1. Setting

Datasets. We conduct experiments on challenging long-video understanding benchmarks. Specifically, we evaluate LensWalk on LVBench [45] and LongVideoBench (long split of validation set) [49] for extended temporal comprehension, and on 3060min split (long split) of Video-MME [13] for general long-video understanding. In addition, we report results on two well-established video reasoning benchmarks, MMVU [67] and Video-MMMU [17], to assess how LensWalk helps on video QA tasks that require higher-level reasoning. We also include EgoSchema [27] to cover egocentric video understanding.

Baselines. We compare against a broad set of methods for video understanding: (1) VLM-based, including both proprietary and open-source vision-language models such as o3 [29], GPT-4.1 [30], Gemini-2.5 Pro [9], and Qwen2.5-VL-72B [2]. (3) Agentic Video Understanding methods, such as VideoAgent [46], VCA [57], MR.Video [32] and

Table 1. Comparison on long video benchmarks. For LensWalk, “(-/-)” denotes Rasoner/Observer; “(M)” means “(M/M)”.

Methods (Reasoner/Observer)	LVBench Overall	LongVideoBench (Val) Long	Video MME Long (w/o sub)	EgoSchema Val
<i>Commercial VLMs</i>				
Gemini-1.5-Pro [42]	33.1	58.6	67.4	-
Gemini-2.0-Flash [14]	48.3	45.7	63.0	71.2
GPT-4o [18]	48.9	60.9	65.3	70.4
GPT-4.1 [30]	51.9	64.6	63.1	72.2
o3 [29]	57.1	60.6	64.7	63.2
GPT-5 [31]	59.8	61.8	68.4	73.8
<i>Open-Source VLMs</i>				
LLaVA-Video-72B [64]	38.7	-	59.6	74.7
InternVL2.5-78B [7]	43.6	-	62.6	-
Qwen2.5-VL-72B [2]	47.7	54.2	63.1	75.4
AdaReTaKe [47]	53.3	-	65.0	-
<i>Video Agents and Others</i>				
VideoAgent [46]	29.3	-	-	63.2
VCA [57]	41.3	-	-	73.6
Ego-R1 [43]	-	-	64.9	68.2
MR. Video [32]	60.8	61.6	61.8	73.0
Deep Video Discovery [62]	74.2	68.6	67.3	76.6
LensWalk (o3/GPT-4.1)	66.8	69.9	70.0	74.8
LensWalk (o3/Qwen2.5-VL-72B)	59.4	67.3	66.7	77.2
LensWalk (o3)	68.6	70.6	71.4	74.8
LensWalk (GPT-5)	66.9	68.8	69.2	74.6

Deep Video Discovery [62]. Most reported results are from official leaderboards or published reports, while we reproduce the results of GPT-4.1 and GPT-5 [31] via official API, with uniformly sampling 256 frames following [32, 62]. More detailed configuration of baselines is in Appendix B.

Implementation Details. As a plug-and-play framework, LensWalk accommodates any combination of reasoning-based language and vision-language models. To demonstrate its potential, we choose combinations of state-of-the-art models such as Qwen2.5-VL-72B, GPT-4.1, o3, and GPT-5 (with minimal reasoning effort, both in baseline and ours), with the Reasoner model also serving as the memory updater by default. Proprietary models are queried via their official APIs. We constrain the agent to a maximum of 20 tool invocations, one per turn. The per-call frame budgets for Scan Search, Segment Focus, and Stitched Verify are set at 180, 32, and 128, respectively, allowing the agent to allocate its observation budget freely within these limits.

4.2. Main Results

As demonstrated in Table 1 and 2, LensWalk achieves strong performance across long video understanding and video reasoning benchmarks. The framework significantly enhances powerful base models, outperforming all *Observer* baselines and multimodal *Reasoners* like o3 and GPT-5 that reason directly over video frames. LensWalk consistently surpasses prior agentic methods, notably achieving over 4% higher accuracy on the challenging VideoMME-Long benchmark than leading approaches like

Mr. Video and Deep Video Discovery, which necessitate a vast number of extra captions to support their reasoning. On the reasoning-focused MMVU and Video-MMMU benchmarks, our framework’s ability to schedule observations for the specialized o3 model boosts accuracy by an additional 2% and 3%. Beyond absolute gains, LensWalk also enables complementary pairing: using o3 as Reasoner over GPT-4.1 observations surpasses either model alone on MMVU, and a “self-play” configuration where a single multimodal reasoning model like o3 serves as both Reasoner and Observer provides strong plug-and-play improvements without any fine-tuning. We report additional shared-model comparisons of other agents and a static extracted-frames baseline in Appendix Table 6, 8.

4.3. Ablation and Analysis

Applicability to Open-Source Models. As shown in Table 3, we validated the framework’s generality by substituting various models as *Reasoner* and *Observer*. The results indicate that LensWalk effectively boosts weaker *Observers*, such as Qwen2.5-VL-7B. Moreover, the open-source *Reasoner* Qwen3-235B-A22B [55] demonstrates a positive impact, notably improving Qwen2.5-VL-7B by 4.3%. However, we observed this open-source *Reasoner* struggles to enhance stronger *Observers* like GPT-4.1 (+0.1%) and Qwen2.5-VL-72B (-0.6%), suggesting its generated observation plans are less effective than these models’ innate reasoning capabilities. This finding highlights the decisive role of the *Reasoner*’s cognitive strength in guiding the acquisition of crucial visual evidence. See more

Table 2. Comparison on video reasoning benchmarks. For LensWalk, “(-/-)” denotes Reasoner/Observer; “(M)” means “(M/M)”.

Methods (Reasoner/Observer)	MMVU		Video-MMMU		
	MC	Percep.	Compreh.	Adap.	Overall
Gemini-1.5-Flash[42]	66.6	57.33	49.00	43.00	49.78
Gemini-1.5-Pro[42]	74.7	59.00	53.33	49.33	53.89
GPT-4o[18]	78.1	66.00	62.00	55.67	61.22
GPT-4.1[30]	76.3	76.00	70.67	55.67	67.44
o3[29]	78.9	79.33	75.67	71.33	75.44
Aria[22]	60.6	65.67	46.67	40.00	50.78
InternVL2-8B[8]	49.1	47.33	33.33	31.67	37.44
Qwen2.5-VL-7B[2]	60.6	58.33	44.33	39.67	47.44
Qwen-2.5-VL-72B[2]	69.3	69.33	61.00	50.33	60.22
LensWalk (o3/GPT-4.1)	80.9	82.00	77.67	71.67	77.11
LensWalk (o3)	79.2	81.33	81.33	72.33	78.33

Table 4. Ablation on the Observation Tools and Reasoning coherence modules of ours(o3/GPT-4.1) on Video-MME (long split).

Observation Tools			Reasoning Coherence		VideoMME Long
Scan Search	Segment Focus	Stitch Verify	Timestamp Anchor	Subject Memory	
	✓	✓			65.4
✓		✓			68.1
✓	✓				66.8
✓	✓	✓	✓		69.7
✓	✓	✓		✓	69.4
✓	✓	✓	✓	✓	70.0

results on more open-source reasoner models in Appendix Table 8

Ablation on Components. We conducted an ablation study to assess the contribution of each component on VideoMME(long split), with results shown in Table 4. The observation tools prove to be highly complementary. Removing *Scan Search*, which is critical for efficiently discovering cues, leads to the largest accuracy drop of 4.6%. The absence of *Stitch Verify*, which impacts the ability to connect disparate scenes for causal analysis, and *Segment Focus*, for extracting fine-grained facts, also degrades performance by 3.2% and 1.9%, respectively. Furthermore, the reasoning coherence modules provide distinct benefits. Adding the *Timestamp Anchor* boosts performance to 69.7%, and incorporating the *Subject Memory* raises it to the final 70.0%, underscoring the importance of both fine-grained temporal alignment and consistent entity tracking in video observation-reason loops.

Efficiency Analysis. To provide a fair comparison of efficiency independent of model deployment or network conditions, we mainly use two key metrics: total token cost and peak per-turn context tokens. As in Figure 3, our reasoning-driven framework avoids the substantial upfront cost of prior methods like Deep Video Discovery (DVD)

Table 3. Results on more Reasoner and Observer choices on VideoMME (long split).

Observer	Reasoner	Acc. (%)
GPT-4.1	-	63.1
	o3	70.0 (+6.9)
	Qwen3-235B-A22B	63.2 (+0.1)
GPT-4.1-mini	-	59.4
	o3	66.0 (+6.6)
	Qwen3-235B-A22B	61.0 (+1.6)
Qwen2.5-VL-72B	-	63.1
	o3	66.7 (+3.6)
	Qwen3-235B-A22B	62.5 (-0.6)
Qwen2.5-VL-7B	-	55.4
	o3	61.3 (+5.9)
	Qwen3-235B-A22B	59.7 (+4.3)

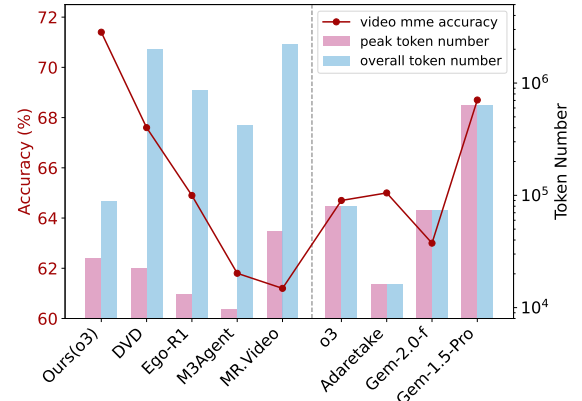


Figure 3. Comparison of video understanding methods on accuracy and token efficiency. Abbrev. of Gemini: Gem.

and Ego-R1 that rely on pre-generating captions for an entire video, much of which is irrelevant. Further comparisons in Appendix (Table 6) show that LensWalk achieves competitive accuracy with substantially fewer frames and much less end-to-end inference time. Furthermore, while consuming total tokens comparable to single-forward baselines, our framework’s reasoning-driven schedule strategically observes across multiple turns. This approach not only enhances performance but also significantly lowers the peak token count per turn, alleviating the severe memory pressure and throughput bottlenecks of processing a massive, monolithic context. In addition, as detailed in Appendix (Table 7), the framework adaptively allocates its observation budget, converging quickly on simple queries while scaling computation for longer or reasoning-intensive cases.

4.4. Analysis on Tool-Call Behaviour

Our design empowers the agent to control its observation scope, enabling it to devise diverse observational paths based on accumulated experience. To better analyze this, we examine the agent’s tool calls and identify distinct behavioral patterns by considering the observation query, the specific tool selected, and the observational scope at each

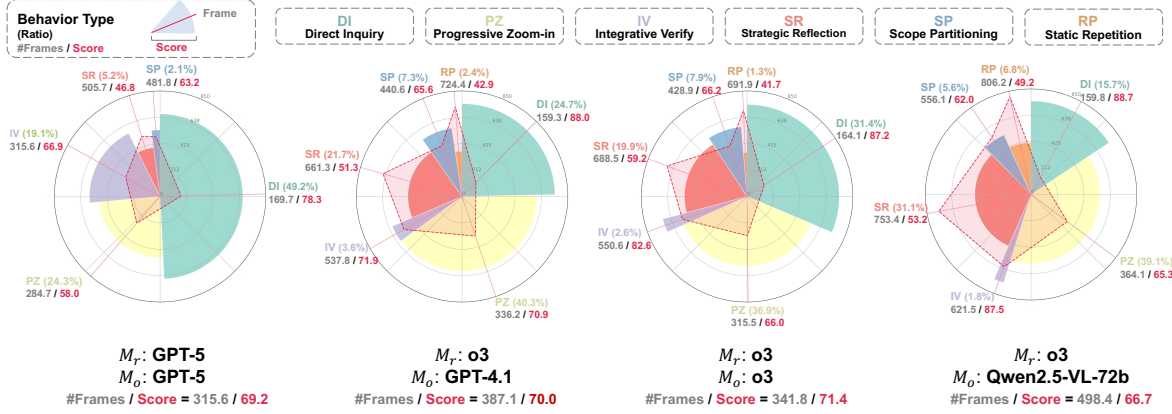


Figure 4. Tool-call behavior of LensWalk across model recipes, visualized following [63]. We group behaviour into six strategy types and show their ratio (sector angle), average frame cost (dashed polygon), and accuracy (noted as score; sector radius), revealing adaptive allocation of more frames to harder queries and diverse exploration under uncertainty.

step. We categorize these patterns into six primary modes (in Figure 4, see Appendix C for details and examples.):

Direct Inquiry. Resolving a task in a single, confident step without prior exploration, typically for high-level or temporally anchored questions.

Progressive Zoom-in. Starting from a broad scan to locate candidate moments and then zooming into promising segments for more detailed inspection.

Scope Partitioning. Partitioning the timeline into distinct scopes, analyzing them independently, and integrating the resulting insights purely at the reasoning level to form a conclusion.

Strategic Reflection. After a focused search hits a dead end, pausing to reassess the approach and launching a new, broader scan over alternative regions or hypotheses.

Integrative Verify. Verifying a complex hypothesis by jointly inspecting multiple key moments in a single stitched observation, enabling holistic confirmation of the conclusion.

Static Repetition. Falling into an unproductive cycle of nearly identical observations over the same region with a static intent, showing little progress or change in strategy.

Analysis of these patterns reveals several key findings:

- **Difficulty-aware cost allocation.** The agent intuitively allocates more observational effort to challenging problems. Hard questions with lower accuracy are associated with longer, more costly observation traces, while simple cases are often resolved efficiently with less frames or even single-step inquiries. This indicates that LensWalk adapts its observation depth based on the evolving uncertainty of each query.
- **Emergent reflective strategies replace inefficient loops.** Empowered by diverse choices in its observation plan, the agent largely avoids the “Static Repetition” behaviour reported in prior work[43, 62]. When faced with uncertainty, it instead invokes reflective patterns such as *Strategic Reflection* or *Integrative Verify* to expand and consolidate evidence. These recovery mechanisms yield more stable and higher accuracy than the rare looping failures that remain.

For a fixed Reasoner, the Observer’s capability is a critical performance driver. A strong Observer (e.g., o3) supplies clear, decisive evidence that directly addresses the query’s intent, enabling frequent, low-cost *Direct Inquiry*. In contrast, weaker Observers (e.g., Qwen2.5-VL-72B) introduce ambiguity, pushing the agent into more complex and expensive observation plans such as retreating to broader scans or repeated verification. This gap highlights that high-quality perception is paramount for cost-effective reasoning.

- **Observer quality shapes strategy and efficiency.** For a fixed Reasoner, the Observer’s capability is a critical performance driver. A strong Observer (e.g., o3) supplies clear, decisive evidence that directly addresses the query’s intent, enabling frequent, low-cost *Direct Inquiry*. In contrast, weaker Observers (e.g., Qwen2.5-VL-72B) introduce ambiguity, pushing the agent into more complex and expensive observation plans such as retreating to broader scans or repeated verification. This gap highlights that high-quality perception is paramount for cost-effective reasoning.

5. Conclusion

In this work, we introduced LensWalk, an agentic framework that actively scheduled video observation based on reasoning in each iteration. By granting a language model the ability to dynamically control what to see, and at what granularity, LensWalk transforms video understanding from a static recognition task into an adaptive reason-plan-observe loop. This design not only yields substantial, plug-and-play performance gains across diverse multimodal models and long-video benchmarks, but also uncovers emergent cognitive behaviors such as progressive zoom-in, reflective replanning, and integrative verification. Our findings highlight that scalable video intelligence lies not only in ever-larger frozen representations, but in scalable visual cognition—where observation is actively orchestrated by reasoning intent. We believe LensWalk provides a concrete step towards the next generation of self-directed multimodal agents, capable of learning how to observe as they think, and ultimately, how to think through seeing.

Acknowledgment

This work is supported by the National Natural Science Foundation of China (Nos.62495084, 62376259).

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